

Autonomous Mobile Robot Navigation System

Abstract

This project simulates an Autonomous Mobile Robot capable of navigating through an environment using occupancy grid mapping, simulated LiDAR sensing, obstacle detection, and A* pathfinding algorithms.

Objective

- Develop autonomous navigation capabilities.
- Build environment maps.
- Plan optimal routes.
- Avoid obstacles dynamically.

Problem Statement

Autonomous robots require the ability to navigate safely in unknown environments while reaching target destinations efficiently.

Software Used

- Python
- NumPy
- Matplotlib

Key Technologies

- Occupancy Grid Mapping
- Simulated LiDAR
- A* Pathfinding Algorithm
- Obstacle Avoidance

Methodology

1. Generate occupancy grid map.
2. Simulate LiDAR sensor readings.
3. Detect obstacles.
4. Apply A* pathfinding.
5. Generate shortest route.
6. Navigate to target location.

Working Principle

The robot scans the environment using simulated LiDAR data and creates a map. A* pathfinding computes the shortest route from the start position to the goal while avoiding obstacles.

Flow of System

LiDAR Scan → Occupancy Grid Mapping → Path Planning → Obstacle Detection → Navigation

Results

The robot successfully generated a collision-free path from the start point to the destination.

Advantages

- Efficient navigation
- Shortest path planning
- Scalable architecture
- Foundation for real-world AMRs

Future Scope

- ROS2 Integration
- Real LiDAR Sensors
- SLAM Implementation
- Dynamic Obstacle Re-planning
- Warehouse Automation

Conclusion

The project demonstrates the core concepts of autonomous navigation, mapping, and intelligent path planning.

References

- ROS Documentation
- A* Pathfinding Research Papers
- Robotics and Autonomous Systems Literature